

Homework 6

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Question 1

a)

As X is uniformly distributed over $[0, 1]$ its pdf is $f(x) = 1$ over that same interval.

$$\mathbb{E}(X) = \int_0^1 x dx = \frac{1}{2}x^2 \Big|_0^1 = \frac{1}{2}$$

$$\mathbb{E}(X^2) = \int_0^1 x^2 dx = \frac{1}{3}x^3 \Big|_0^1 = \frac{1}{3}$$

Therefore:

$$Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

b)

$$M_X(t) = \int_0^1 e^{tx} dx = \frac{e^{tx}}{t} \Big|_0^1 = \frac{e^t - 1}{t}$$

Question of course is, what happens when $t = 0$? Let's use L'Hopital's rule by taking the derivative of both numerator and denominator:

$$\frac{e^t}{1}$$

At $t = 0$ this will be 1 and so $t \rightarrow 0$ of $\lim_{t \rightarrow 0} M_X(t) = 1$. We're good to use this formula.

c)

i.

$$\begin{aligned}\mathbb{E}(Y) &= \mathbb{E}(a + (b - a)X) = \mathbb{E}(a) + \mathbb{E}((b - a)X) = a + (b - a)\mathbb{E}(X) \\ &= a + (b - a)/2 = \frac{a + b}{2}\end{aligned}$$

$$\begin{aligned}\mathbb{E}(Y^2) &= \mathbb{E}((a + (b - a)X)(a + (b - a)X)) = \mathbb{E}(a^2 + 2a(b - a)X + (b - a)^2 X^2) \\ &= \mathbb{E}(a^2) + 2a(b - a)\mathbb{E}(X) + (b - a)^2\mathbb{E}(X^2) = a^2 + a(b - a) + \frac{(b - a)^2}{3}\end{aligned}$$

$$\begin{aligned}Var(Y) &= \mathbb{E}(X^2) - \mathbb{E}(X)^2 = a^2 + a(b - a) + \frac{(b - a)^2}{3} - \frac{(a + b)^2}{4} \\ &= \frac{12ab + 4(b^2 - 2ab + a^2) - 3(a^2 + 2ab + b^2)}{12} = \frac{(b - a)^2}{12}\end{aligned}$$

ii.

$g(x) = a + (b - a)x$ is monotonic, invertible ($g^{-1}(y) = (y - a)/(b - a)$), and has a continuous derivative. So:

$$f(y) = f(g^{-1}(y)) \left| \frac{dx}{dy} \right| = \frac{1}{b - a}$$

For $y \in [a, b]$.

We have ourselves a uniform distribution over $[a, b]$.

iii.

It's just so easy to use the pmf:

$$M_Y(t) = \int_a^b \frac{1}{b - a} e^{ty} dy = \frac{e^{ty}}{(b - a)t} \Big|_a^b = \frac{e^{bt} - e^{at}}{(b - a)t}$$

Question 2

a)

$g(x) = \sigma X + \mu$ is monotonic, invertible ($g^{-1}(y) = (y - \mu)/\sigma$), and has a continuous derivative. So:

$$f(y) = f(g^{-1}(y)) \left| \frac{dx}{dy} \right| = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} \frac{1}{\sigma} = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

b)

$$M_Y(t) = e^{\mu t} M_X(\sigma t) = e^{\mu t} e^{(\sigma t)^2/2}$$

c)

$$M'_Y(t) = e^{\mu t + \sigma^2 t^2/2} (\mu + \sigma^2 t)$$

$$M''_Y(t) = e^{\mu t + \sigma^2 t^2/2} (\mu + \sigma^2 t)^2 + \sigma^2 e^{\mu t + \sigma^2 t^2/2}$$

$$\mathbb{E}(Y) = M'_Y(0) = \mu$$

$$\mathbb{E}(Y^2) = M''_Y(0) = \mu^2 + \sigma^2$$

$$\text{Var}(Y) = \mathbb{E}(Y^2) - \mathbb{E}(Y)^2 = \sigma^2$$

Question 3

$$M''_X = e^{t^2/2} t^2 + e^{t^2/2}$$

$$M'''_X = e^{t^2/2} t^3 + 3e^{t^2/2} t \rightarrow \mathbb{E}(X^3) = 0$$

$$M''''_X = e^{t^2/2} t^4 + 6e^{t^2/2} t^2 + 3e^{t^2/2} \rightarrow \mathbb{E}(X^4) = 3$$

Question 4

$$f(x) = \frac{1}{\Gamma(a)b^a} x^{a-1} e^{-x/b}$$

where $x \geq 0$

$$f'(x) = C \left((a-1)x^{a-2}e^{-x/b} - \frac{1}{b}x^{a-1}e^{-x/b} \right) = Cx^{a-2}e^{-x/b} \left((a-1) - \frac{x}{b} \right)$$

where $C = \frac{1}{\Gamma(a)b^a}$ is a positive constant.

Therefore if a maximum is going to occur it will occur either at $x = 0$ (one end of our range) or where $f'(x_{inflect}) = 0 \rightarrow x_{inflect} = b(a-1)$. If it doesn't occur at one of these two places there is no maximum.

a)

Because $a < 1$, $x_{inflect} < 0$ which is not part of our range. So the maximum occurs at 0 or it doesn't occur at all. Question is which sign does our derivative have? Is $f(x)$ increasing at 0 or decreasing? Our sign is determined by:

$$\left((a-1) - \frac{x}{b} \right) \rightarrow (a-1)$$

which is less than 0 because $a < 1$. Therefore $x = 0$ has the maximum value of $f(x)$ and is our mode.

b)

Here $x_{inflect} = 0$. Also we note that in this case:

$$f(x) = \frac{1}{\Gamma(1)b} e^{-x/b}$$

So the function is most certainly decreasing as x increases. Therefore, once again, our mode is 0.

c)

In this case $x_{inflect} = b(a-1) > 0$ so we need to evaluate the second derivative at that point.

$$\ln f' = \ln C + \ln(x^{a-2}) + \ln(e^{-x/b}) + \ln(a-1-x/b)$$

$$\frac{f''}{f'} = \frac{(a-2)x^{a-3}}{x^{a-2}} + \frac{-e^{-x/b}}{be^{-x/b}} + \frac{-1}{b(a-1-x/b)}$$

$$f'' = f' \frac{a-2}{x} - f' \frac{1}{b} - f' \frac{1}{b(a-1-x/b)}$$

Only the last term is going to cancel out the bit of f' that is driving it to zero at $x = b(a-1)$ so it all comes down to that last term which is:

$$-Cx^{a-2}e^{-x/b} \left((a-1) - \frac{x}{b} \right) \frac{1}{b(a-1-x/b)}$$

which has to be negative. Therefore our inflection point $x = b(a-1)$ is indeed a maximum and therefore a mode!

Question 5

$g(x) = kX$ is monotonic, invertable ($g^{-1}(y) = y/k$), and has a continuous derivative. So:

$$f(y) = f(g^{-1}(y)) \left| \frac{dx}{dy} \right| = \frac{1}{\Gamma(a)b^a} (y/k)^{a-1} e^{-y/kb} \frac{1}{k}$$

$$= \frac{1}{\Gamma(a)(kb)^a} y^{a-1} e^{-y/kb}$$

Therefore Y is distributed as $Gamma(a, kb)$.

Question 6

a)

$$M_X(t) = \sum_{x=0}^{\infty} e^{xt} \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t-1)}$$

b)

$$M'_X(t) = e^{\lambda(e^t-1)} \lambda e^t \rightarrow \mathbb{E}(X) = \lambda$$

$$M''_X(t) = e^{\lambda(e^t-1)} (\lambda e^t)^2 + e^{\lambda(e^t-1)} (\lambda e^t) \rightarrow \mathbb{E}(X^2) = \lambda^2 + \lambda$$

$$Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \lambda$$